

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 4th Year 2nd Term Examination-2022
LE 4213
Testing of Leather and Allied Materials

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | Discuss the motives of conducting physical testing. | 05 |
| 1(b) | Suppose you have been provided with a truck of 280 pcs of leather specimens. How will you select the specimens for physical test and chemical tests? Illustrate a report while the specimens pass the thumb tests for upper leather or not. | 20 |
| 1(c) | Explain the sampling position from bend and butt portion of a leather. | 10 |
| 2(a) | Discuss the significance of strength and stretch of a leather specimen. | 08 |
| 2(b) | "Percentage of elongation for a leather specimen depends on the applied force through it"- Explain the statement. | 09 |
| 2(c) | "Tensile strength of a chrome tanned leather decreases appreciably when that is heavily retanned with vegetable tannins"-Explain. | 11 |
| 2(d) | Explain the role of moisture and fat content in affecting the tensile strength of leather. | 07 |
| 3(a) | "Partial removal of inter-fibrillary materials from hides during liming improve the tensile strength of the finished leather but decrease the stitch tearing strength"-Explain the statement. | 10 |
| 3(b) | Mention the standard values of different fiber strength tests with proper units. | 06 |
| 3(c) | Why sampling preparation and sampling direction are important prior to physical testing? | 08 |
| 3(d) | Explain the mechanism of elasticity test of finished film and its controlling factors. | 11 |
| 4(a) | How light fastness of finished film and water fastness properties of leather can be improved? | 09 |
| 4(b) | The bursting strength is an index of the overall strength of leather-Explain. Compare between tensometer and lastometer. | 12 |
| 4(c) | Discuss the tearing mechanism of leather. Show that "leather made up of less elastic tanned collagen fibers, can show high elongation". | 14 |

SECTION-B

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|------|---|----|
| 5(a) | Define Fick's law of diffusion. Mention the unit of water vapour permeability and explain it. What's the standard value of WVP for shoe upper and lining materials? | 12 |
|------|---|----|

5(b)	Briefly discuss the scope, principle and assessment process of Water Vapour Permeability test.	13
5(c)	Mention the expression of result of following test: (i) Scuff resistance test and (ii) Softness test	06
5(d)	Why high scuff resistance is needed for kiddy shoe leather?	04
6(a)	How to prepare a perspiration solution to carry out the perspiration resistance test of leather?	04
6(b)	State the principle and standard procedure for the determination of softness test of gloving leather.	10
6(c)	Cite the importance of using different shapes of the ring during the softness test of leather.	07
6(d)	The shrinkage temperature of any finished leather provides which information to the users.	06
6(e)	Why dome plastometer test is important? What's the dimension of this test specimen?	08
7(a)	What is Bennewart flexing? Why this is performed? State the test methodology, inspection state and standard requirements for soling materials.	14
7(b)	Why adhesion test is conducted? State the sequential operation and standard requirements for different soling materials.	12
7(c)	What is the Ford Abrasion test? State the standard operating procedure to determining scuff resistance of upper leather.	09
8(a)	Define fasteners and accessories. Mention the importance of fasteners and accessories assessment for footwear products manufacturing.	07
8(b)	Write short notes on: (i) Attachment strength of slide fasteners and (ii) Strength test of eyelet to a shoelace.	12
8(c)	Why maser test is done? Mention the dimension of the test specimen of this test.	06
8(d)	Illustrate the influencing factors for the fogging value of upholstery leather manufacturing.	10

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 4th Year 2nd Term Examination-2022
LE 4025
Materials Science for Footwear Products

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

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iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Classify man-made fibers that are used for producing shoe upper materials. How are fabrics produced from fibers? 07
- 1(b) "Poromeric shoe upper is developed in a different way compare to others"-briefly explain the manufacturing process of poromerics. 07
- 1(c) Describe different types of interlacing methods in woven fabrics, including their properties. 09
- 1(d) Write short notes on: (i) Computer controlled 4D knitting (ii) Braided fabrics and (iii) Thermoplastic Polyurethane shoe upper. 12
- 2(a) What is meant by halogenation? 04
- 2(b) Define the following adhesive terminologies: (i) Drying time (ii) Activation temperature and (iii) Creep strength. 12
- 2(c) Make a comparison between latex and rubber solution adhesives based on the ingredients and properties. 07
- 2(d) "Some of the adhesives do not require any solvent for application". Briefly describe these kinds of adhesives that are used in footwear industries. 12
- 3(a) Convert the following needle size (American to European and Vice-Versa): (i) 8 (ii) 75 (iii) 80 (iv) 11 09
- 3(b) Explain briefly about the ingredients used in PVC shoe sole manufacturing. 15
- 3(c) Define Reaction Injection Moulding (RIM). "PU shoe sole can be classified in different ways". Discuss briefly the classification. 11
- 4(a) How does a phylon sole production process differ from an EVA sole? 06
- 4(b) Why is two-phase structure possible in thermoplastic rubber sole? State the pros and cons of TPR sole. 12
- 4(c) What is meant by vulcanization? "Vulcanized rubber soles can be produced in different ways". Illustrate this through the flow chart. 08
- 4(d) Write short notes: (i) Microcellular rubber sole (ii) Slip-resistant sole and (iii) Injected EVA 09

SECTION-B

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|------|---|----|
| 5(a) | Why toe puff is used? Write about celluloid-impregnated fabric as toe puff. | 10 |
| 5(b) | Define toe puff. What is the working and wearing properties of an ideal toe puff? | 08 |
| 5(c) | How strap is reinforced? | 05 |
| 5(d) | Write short notes on: (i) Cavity heel (ii) Shoe dressing and (iii) Touch and close fastener. | 12 |
| | | |
| 6(a) | Draw the flow chart for insole manufacturing for footwear assembling. | 10 |
| 6(b) | How can you identify a cellulose insole board? What are the pros and cons of cellulose board? | 07 |
| 6(c) | Define shank. "Shank works as a bridge between insole and sole"-Explain. | 06 |
| 6(d) | Write about different sorts of steel shank. What are attributes of steel shank over wooden shank? | 12 |
| | | |
| 7(a) | Briefly explain the methods of shoe lacing which are frequently used. | 15 |
| 7(b) | Mention the different parts of an eyelet. Compare among stared, ordinary and round settings of the eyelet with related figures. | 10 |
| 7(c) | How direct reinforcement is done and why? | 04 |
| 7(d) | Distinguish between eyelet and grommet. | 06 |
| | | |
| 8(a) | Classify waxes that are used as shoe finishes. Why is carnauba wax widely used as a shoe-finishing material? | 10 |
| 8(b) | Expound briefly about the insole lip with necessary figures. | 07 |
| 8(c) | Briefly discuss the different types of bottom fillers that are used in footwear product manufacturing. | 10 |
| 8(d) | "Performance of touch-and-close fastness largely depends on the accuracy of attachment". Explain it. | 08 |

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 4th Year 2nd Term Examination-2022
IPE 4219
Industrial Management and Entrepreneurship

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

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iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Define management. Draw a network diagram of management functions and explain it. 13
- 1(b) Define managerial skills. Describe the necessary skills that have to be possessed by a manager in brief. 12
- 1(c) Define span of management. Explain different types of superior-subordinate relationships. 10
- 2(a) What do you mean by organizational structure? Describe line and staff organization with necessary figures. 12
- 2(b) What is centralized organizational structure? Explain decentralization with its advantages and disadvantages. 13
- 2(c) Write down the barriers to successful communication and possible outcomes to overcome those barriers. 10
- 3(a) What is motivation? Discuss the levels of involved in Maslow's Need Hierarchy theory. 13
- 3(b) Define group dynamics. Write down the assumptions of Theory -Y. 10
- 3(c) Why delegation of authority is necessary? Discuss the obstacles of effective delegation. 12
- 4(a) Define recruitment. During recruitment process, how are the most suitable candidates selected? 10
- 4(b) What is conflict? Describe the preventive measures of organizational conflict. 10
- 4(c) A job is rated in terms of wages TK 900 per day. The standard time set for the job is 16 days, 8 hrs./day. Three workers have taken 112, 120 and 140 hrs, respectively for the completion of the job. A bonus of 75% on the time taken will be given only to those who have completed the job in the standard time. Calculate the earnings of each individual by Hasley Plan. 15

SECTION-B

- 5(a) Define entrepreneurship. Elon Musk is the CEO and co-founder of Tesla and Sundar Pichai is the CEO of Alphabet Inc. (parent company of Google). Even though they are both CEOs, their roles in their respective companies are quite different. What are the differences in their roles? Justify your answers. 13

- 5(b) Suppose you are a newcomer entrepreneur in the leather shoes market. You don't have huge capital on your hand. In terms of product planning, what strategy or behavioral choice should be applicable in your situation? Should you continuously upgrade the products or stick to your backdated products? 12
- 5(c) What kinds of people are motivated to be entrepreneurs because they feel the need for affiliation? 10
- 6(a) How does a successful manager create a balance between effectiveness and efficiency? Explain with suitable scenarios when effectiveness will be desirable and when efficiency is needed. 12
- 6(b) What is ambidexterity? In terms of strategic management how would you conduct SWOT analysis for a company like Apex? 12
- 6(c) Briefly discuss about various characteristics of MIS. 11
- 7(a) Explain about different types of partners in the partnership organization. 12
- 7(b) How does an LLP differ from a company? Make the necessary distinctions among their features. 13
- 7(c) In 2007, Netflix started streaming services for movies and tv shows which allowed the customers to access a large library of contents without physical discs. What types of innovation is this in terms of hi-tech entrepreneurship? 10
- 8(a) What is technology transfer? Discuss the models of transfer briefly. 12
- 8(b) Why did companies like Nokia and Kodak fail but their competitors like Samsung and Sony succeed even though they had less market share than Nokia and Kodak? 10
- 8(c) Differentiate between communication-driven, data driven, knowledge-driven and model-driven DSS. What are some of the applications of DSS? 13

Department of Leather Engineering
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 B. Sc. Engineering 4th Year 2nd Term Examination-2022
LE 4207
Dyeing & Finishing II

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

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iii) Assume reasonable data if any missing.

SECTION-A

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|---|----|
| 1(a) State and explain the necessity of adsorption isotherm. | 08 |
| 1(b) Illustrate different categories of adsorption isotherms, their applications with significances. | 10 |
| 1(c) How is it possible to understand about adsorption isotherm of an adsorbate through SEM images? | 05 |
| 1(d) Explain how Langmuir isotherm model can reflect a hyperbolic pattern. | 12 |
| 2(a) “There remains no lateral interaction between the adsorbed and localized molecules”. Explain with respect to Langmuir isotherm model. | 08 |
| 2(b) How the nature of physisorption can be understood in an equilibrium state? -explain mathematically. | 15 |
| 2(c) Write down the significance of dye aggregation in completion of dyeing process. | 06 |
| 2(d) Explain the graphical representation of Freundlich isotherm model. | 06 |
| 3(a) Explain the mechanism of dyeing with necessary diagram. | 08 |
| 3(b) Discuss the mechanism of micelle formation of dye molecules and its significance. | 12 |
| 3(c) Suppose you have been provided with 200 semi-spherical shaped dye molecules which are two times wider than their widths and having a viscosity of $0.00287 \text{ mgm}^{-1}\text{min}^{-1}$. Calculate the molecular size of the molecules of diffusion co-efficient of $3.11 \times 10^{-7} \text{ cm}^2/\text{min}$ at 95°F . | 15 |
| 4(a) “Diffusion co-efficient of dye molecule depends on their concentration in a closed system”. Explain with figure. | 10 |
| 4(b) “Aggregation number of colloidal particles depends on the density of solvent those are dissolved in”. Derive the statement mathematically. | 08 |
| 4(c) What is the significance of chemical kinetics in adsorption process? | 08 |
| 4(d) Derive the equation of “Heat of dyeing” calculation. | 09 |

SECTION-B

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|---|----|
| 5(a) What are the key drivers for the adaption of waterborne coating for industrial applications? | 05 |
|---|----|

5(b)	“The application process of powder coating is different from other coating processes”. Explain the statement.	10
5(c)	How the rheological process is associated with powder coatings?	05
5(d)	Describe the mechanism of environmental pollution caused by VOCs and their possible impacts.	10
5(e)	Define the following terms: i) Flooding and ii) Floating.	05
6(a)	Explain the film formation mechanism by using Williams-Landel-Ferry equation.	10
6(b)	Describe the film formation mechanism of cross-linking polymers.	10
6(c)	State the solvent removal mechanism of thermoplastic coating.	07
6(d)	What are the problems associated with the storage of coatings and probable ways to solve those problems?	08
7(a)	How can we observe secondary color although we have dedicated cone cells for primary colors?	10
7(b)	Explain the best suited color mixing model for the mixing of pigments and dyes to get desired color.	07
7(c)	Why do two objects appear to match in color under one lighting condition but not under another?	08
7(d)	State the significance of the Kubelka-Munk analysis for leather coating.	10
8(a)	Describe the leather surface coating steps with specific purposes.	07
8(b)	How do color matching cabinet help to ensure accurate color matching?	08
8(c)	Provide a comprehensive overview of water fastness test and plasticizer resistance test including their objectives and methodologies.	08
8(d)	Classify color blindness according to the Young-Helmholtz theory of color vision.	05
8(e)	Which color will you observe if you see a magenta-colored substance under yellow light? -Describe.	07

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 B. Sc. Engineering 4th Year 2nd Term Examination-2022
LE 4217
Production and Quality Control

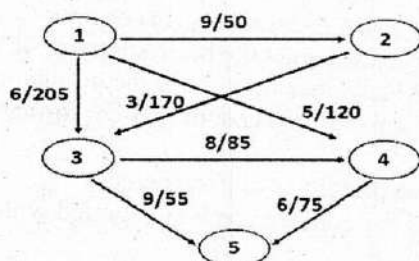
Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
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SECTION-A

- 1(a) Define job production. Explain the phases of product life cycle in a particular market. 10
 1(b) Write down the key steps for improving productivity. 10
 1(c) Why forecasting is called the art and science of predicting future events. 05
 1(d) An automated guided vehicle system has the following data for load and unload times are 25s each and speed of AGV = 1 m/s, traffic factor = 0.8. How many vehicles are required to satisfy a delivery requirement of 28 del/hr.? Assume, $A = 0.96$, $E = 1$. 10



- 2(a) Define sales forecasting. Write down the importance of sales forecasting. 10
 2(b) What is time series method of forecasting? Explain the components of time series method of sales forecasting. 10
 2(c) Given below are the sales pattern of ABC footwear company, manufacturing consumer goods. Trends of consumption of consumer goods of similar goods in the market is also depicted. Find out the correlation between sales of the company and consumption of goods in the market using standard deviation method when the consumption in the market will rise to 25 lakhs. 15

Sales (Rs. Lakhs)	Consumption of goods in the market (Rs. Lakhs)
X	Y
1	9
2	6
3	10
4	14
5	15
6	13
7	12
8	16
9	11

- 3(a) State the importance of production planning and control systems for products industry. 08
 3(b) What is master production scheduling (MPS)? BD Footwear limited produces footwear for consumers and other manufacturers on a produce-to-stock basis. The production planner is developing an MPS for footwear for the next 6 weeks. The minimum lot size is 1500 pairs; safety stock level is 400 pairs. There are currently 1120 pairs in inventory. The estimates of demand for footwear pair in the next 6 weeks are shown below: 10

	Week					
	1	2	3	4	5	6
Customers	500	1000	500	200	700	1000
Other Manufacturers	200	300	400	500	300	200
Market research	0	50	0	0	10	0
Production research	10	0	0	0	0	0

- Develop a master production schedule for footwear for the next 6 week.
 3(c) What are the options available for managers to meet the fluctuating future demand? 09

- 3(d) Describe the open loop and closed loop system of production control. 08
- 4(a) What is the law of variable proportion? Discuss the steps of production based on the law of variable proportions with suitable example. 15
- 4(b) Business data of "Z" footwear company are given below: 20

Time horizon	Jan	Feb	Mar	Apr	May	Jun	Total
Demand forecasting (in pairs)	1700	1500	1200	1000	1300	1400	8100
Working days	25	22	25	24	23	24	143

Where, beginning inventory 300 pairs, beginning labor 40 workers, labor time required 0.3 unit/hr, inventory holding cost \$1.5 /U/M, back orders \$4.0 /U/M, hiring and firing cost \$ 200/workers, lay off cost \$ 300/workers, straight time cost (8 hr) \$ 5/hr, outsourcing cost \$ 30/U. If the company follows the level strategy of production planning, then calculate the total cost of this strategy.

SECTION-B

- 5(a) Clarify the terms: (i) Quality, (ii) Quality Assurance, (iii) Quality Control and (iv) Q.M. S. 08
- 5(b) What is meant by House of quality? Construct the basic structure of House of quality. 12
- 5(c) What is meant by quality trilogy? How does it work for the product quality optimization? 10
- 5(d) Distinguish between cost of conformance and cost of non-conformance. 05
- 6(a) What is Zero defects? Mention the principles of Zero defects. 07
- 6(b) Illustrate Kano Model. How does the concept can be integrated with leather products design and development? Explain with example. 12
- 6(c) State the Six-Sigma concept. Describe the tools of quality improvement towards zero defects in the footwear industry. 16
- 7(a) Write down the differences between quality control and quality assurance. 04
- 7(b) PERT and CPM are complementary tools of project management- explain with example. 06
- 7(c) Define critical path of a project. The fabrication process of a wallet is scheduled in a way that it can take 23 minutes to fabricate. The following parameters are associated with the process: 25

Activity	O _T	M _T	P _T	Predecessors
A	2	8	14	-
B	4	4	4	-
C	4	10	16	A
D	6	8	10	A
E	8	12	16	C, B
F	1	1	12	C, D
G	6	12	18	D, E

Draw the network diagram and find (i) expected time of activities, variances, and slacks (ii) CPM (iii) Z-score to remark on probable fabrication time.

- 8(a) What is meant by EOQ? How does it provide the minimum quantity that grabs the holding cost along with ordering cost? Explain with example. 08
- 8(b) Define EIA. Describe the components of EIA. Describe Gantt chart with application. 20
- 8(c) The demand of a leather jacket is 30000 pics/year. One unit of the item costs US \$220, holding cost rate 15% / year. Setup time to produce a batch is 5.0 hr. Cost of equipment plus labor is US \$110 / hr. Determine EOQ for this case. 07